Business Problem Statement: Product Review "Sentiment & Action" Analytics

Organizations receive large volumes of product reviews from e-commerce platforms, app stores, and customer surveys. While these reviews contain valuable insights into customer satisfaction, pain points, and product issues, extracting meaningful and actionable intelligence manually is inefficient, inconsistent, and not scalable.

A product analytics team aims to build an intelligent system that can automatically analyze customer feedback, detect sentiment patterns, identify recurring themes, and highlight the most critical issues affecting user experience. The goal is to transform unstructured review data into structured insights that can drive product improvements and strategic decision-making.

You are tasked with developing an end-to-end analytics solution to address the following business question:

"How can large-scale product review data be leveraged to automatically detect sentiment, uncover key customer pain points, and generate actionable product improvement recommendations in a scalable and intelligent way?"

Deliverables

1. Data Engineering & Modeling (Python)

- Ingest and preprocess product review data (CSV files or APIs)
- Clean and filter noisy data (e.g., spam or bot-generated reviews)
- Generate text embeddings for reviews
- Build and evaluate sentiment classification models
- Apply clustering techniques (vector clustering + LLM-based grouping) to identify themes
- Develop an LLM-powered agent to suggest product improvements based on clustered insights

2. Data Storage & Querying (SQL)

- Store engineered datasets and model outputs in a structured database
- Design schema for macro indicators, features, and regime timelines
- Enable efficient querying for regime trends, transitions, and comparisons

3. Visualization & Insights (Streamlit Dashboard)

- Develop an interactive dashboard using Streamlit to visualize:
- Sentiment distribution (positive, negative, neutral)
- Key themes and topic clusters derived from review embeddings
- Top customer pain points and frequently occurring issues
- Sentiment and topic trends over time (weekly/monthly analysis)

4. API Development (FastAPI)

- Develop RESTful APIs using FastAPI to serve model predictions and analytics
- Create endpoints for:
    - Uploading or fetching product review data
    - Performing sentiment analysis on new reviews
    - Retrieving clustered themes and insights
    - Generating product improvement recommendations
- Ensure efficient request handling and low-latency responses
- Implement input validation and error handling

- Enable integration between the Streamlit dashboard and backend services

5. AI & Intelligence Layer: Build and integrate AI components for advanced text analytics

- Text embeddings generation using transformer-based models
- Sentiment classification using NLP models
- Theme extraction using clustering (e.g., KMeans, vector similarity)
- LLM-based agent to generate actionable product improvement suggestions